

Armaan Jass

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EDUCATION

University of Guelph

Sept 2024 – April 2029

Bachelor of Engineering, Mechanical Engineering

Guelph, ON

- Relevant Coursework: Eng Design I–II, Materials Science, Mechanics I–II, Machine Design, Thermodynamics
- Dean's Honour List (Winter/Fall 2025, Winter 2026); Overall GPA: **3.7/4.0**

WORK EXPERIENCE

Hydraulic/Electrical Technician, Angus Irrigation

[View Work Details](#)

Hamilton, ON

- Troubleshoot, installed, and repaired hydraulic and electrical irrigation components by tracing wiring, controllers, solenoids, valves, and piping networks to systematically isolate faults and determine practical repairs.
- Calculated property water requirements and zone flow capacities to establish watering schedules while sizing controllers and field wiring to exact zone demands and planning piping routes to reduce unnecessary **material use and labour**.
- Diagnosed leaks and intermittent system failures through observation, testing, and logical elimination, prioritizing accurate electrical diagnostics before excavation to avoid unnecessary work and protect **customer costs and project timelines**.
- Interpreted blueprints, site plans, and existing system layouts to identify piping routes, valve locations, and control zones while producing field sketches to document system modifications and support future troubleshooting.

Project Manager, Enabled Talent

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Remote

- Conducted a market feasibility study across target organizations in Ghana to identify accessibility needs and employment opportunities for individuals with disabilities and out-of-school youth.
- Researched local labour market demand across companies, startups, universities, and community organizations to identify viable partnership and outreach targets while accounting for limited technology access.
- Designed an outreach and onboarding strategy incorporating community-based iPad stations and built a phased project timeline covering research, outreach, and onboarding activities.
- Evaluated staffing, travel, and equipment costs and delivered a comprehensive **30+ page** feasibility and scoping report with research data and supporting tables to executive leadership.

ENGINEERING PROJECTS

Autonomous Multi-tool Drone, Personal Project

[View Project Details](#)

Cambridge, ON

- Designed and iterated drone housing structures in Fusion 360 using **PA6-CF**, applying machining and thermal heat-set inserts to improve **structural rigidity** and component integration while achieving an **8% weight reduction**.
- Upgraded the power system from **4S (14.8 V) to 6S (22.2 V)** LiPo, increasing nominal voltage by **50%** to provide consistent motor power, synchronize motor speeds, and increase available **thrust capacity**.
- Calibrated, tested, and validated flight parameters using QGroundControl and Mission Planner while integrating the Pixhawk 6C, 4-in-1 ESC, GPS, ELRS receiver, Raspberry Pi 4, and vision sensor for autonomous operation.

Brakes, Aerodynamics and Driveline Member, Formula SAE

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University of Guelph, ON

- Reverse engineered a mechanical throttle body using precision measuring tools and SolidWorks, producing and validating a functional 3D-printed assembly for vehicle testing and driving.
- Applied engineering judgment to select a **lower-cost, lower-weight** 3D-printed polymer alternative to metal components while considering operating temperature, heat transfer, durability, and application requirements.
- Engineered brake fluid reservoir mounting hardware using **DFM/DFA** principles while accounting for vibration loading, fastener access, assembly sequence, serviceability, and brake line routing.
- Supported vehicle fabrication through welding and composite manufacturing, including fiberglass and carbon fiber hand layup, while preparing dimensioned 2D drawings with **GD&T** tolerance callouts.

Design II – Mechanical Engineering Project

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University of Guelph, ON

- Collaborated in a team to reverse engineer a physical toy assembly into SolidWorks, modeling **50+ CAD components** and creating multiple subassemblies, motion simulations, and dimensioned 2D engineering drawings.
- Translated physical measurements into parametric 3D models and functional assemblies, applying **tolerancing** and design-for-assembly considerations to integrate individual components into the final product.

LEADERSHIP & INVOLVEMENT

Team Captain, Provincial Soccer

2023 – Present

Oriental Sports Club - Cambridge, ON

- Led team strategy, tactical planning, and squad communication under high-pressure competitive environments, driving performance to secure a regional championship.